Ad Omnia

Ad Omnia in perpetuum.

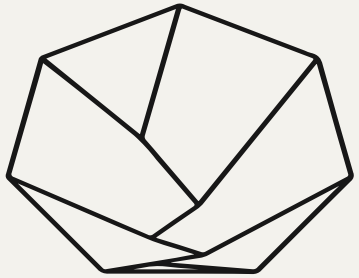


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Now with the support of:



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A short recap



prediction analysis control

GDPR - What Data, and Why

Station & operator data
Station ID + license hash control who can act on the platform; every change is attributed, for accountability.

Vehicle & crew positions
Shown to coordinate resources on the map - never used to evaluate or rank a crew's performance.

Incident, order & offence data
The operational picture itself. Offence data from police feeds stays under official authority only (Art. 10 GDPR).

Victim data
Arrives already anonymised from INEM's API - so it isn't personal data for Ad Omnia at all.

The Erasure Paradox

The Conflict

NFR-INT-1: no audit, order or decision record can ever be edited or deleted. Art. 17 GDPR gives people a right to erasure.

Why It's Lawful

Erasure doesn't apply when the data is needed for a legal obligation or a public task (Art. 17(3)(b)) - emergency decision logs are exactly that.

What the design still does

- Audit entries store IDs and field names only - never the text of a description or an order.
- Personal free text lives in the operational tables, where it can be erased once its retention period ends.
- Operators are identified by station ID - pseudonymous on its own; the person-to-station mapping stays in the agency's own HR systems.



Security & Sharing

Security Measures

- Licenses stored with a slow, salted hash - never in plain text.
- Encryption in transit (TLS) and at rest, before go-live.
- One-way flow: data moves private to public only, nothing flows back in.
- Plugins run sandboxed, on an internal-only network.

Sharing & Roles

- Ad Omnia acts as processor; each agency is controller of its own data.
- Shared with other agencies only inside the joint operational picture.
- The public app receives validated incidents only - never confidential data.
- No transfer outside the EU, under any circumstance.

PoC today: public, synthetic & test data only. GDPR applies in full once a real pilot uses agency data.

Intellectual Property

Today: academic project

Rule in force	UA IP Regulation (Senate, 2001)
Applicable point	1.2: work using UA means and resources
Rights holder	University of Aveiro
Code license	Proprietary, all rights reserved

If we commercialize

1. Inform the University
Points 4 and 5 of the regulation

↓

2. University decides
Within 60 days

↓ ↓

Releases rights
Team gets all rights (5.2)

Keeps rights
Creators share the benefits (8)

Third-Party Licenses

React, MapLibre GL JS, FastAPI, LightGBM	MIT / BSD
Kafka, gRPC, OR-Tools, Docker Engine	Apache-2.0
Python	PSF-2.0
PostgreSQL	PostgreSQL License
TimescaleDB	Apache-2.0 + Timescale License
Geist, JetBrains Mono (fonts)	SIL Open Font License

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license conflicts

All

allow commercial use



Data Source Licenses

Source	Used for	Main condition	Commercial use	Decision
Fogos.pt	Real-time wildfires	Token granted; no ads or sales	Not allowed	Authorized
IPMA	Warnings, fire risk, weather	Free use, no profit; logo and link	Not allowed as written	Use, with attribution
OpenSky	Aerial assets (ADS-B)	Research, education, internal use	Ask OpenSky	Use for the PoC
Open-Meteo	Weather history, forecast	CC BY 4.0; free API non-commercial	Paid plan	Use for the PoC
dados.gov.pt	Zone boundaries, history	CC BY 4.0 by default, per dataset	With attribution	Record each dataset
Maps	Basemaps	Self-hosted; ODbL and others	With attribution	Self-host
OurAirports	Airport reference data	Public domain	Allowed	Use freely



What the data terms require

Commercial Use

- **Fogos.pt, IPMA**: not allowed as written
- **OpenSky**: replace or license
- **Open-Meteo**: paid plan or self-hosted
- **Other sources**: open, with attribution

Attribution

- **IPMA**: logo with link
- **Fogos.pt, OpenSky**:
- "Source: …"
- **Open-Meteo**: CC BY 4.0 credit
- **Maps**: Protomaps, OSM and others

Personal data

- OpenSky (ICAO24, callsigns) and Fogos.pt (OSINT text) can touch personal data
- We do not identify individuals
- The public app does not collect personal data



What the users agree to:

Not a replacement for 112

Ad Omnia is not to be used as a replacement for established emergency communication services.

Full data accuracy is not guarenteed

Since data is ingested from external sources, its accuracy and time-sensitivity cannot be fully expected by the user.

The app never decides

Predictions and recommendations are not a replacement for human actors. They are to be evaluated by competent individuals before being acted upon.

Does not encompass the totality of the Terms and Conditions.

Accessibility

Severity rings around the incident



Low · Moderate · High · Critical

Resource status pattern on the bar



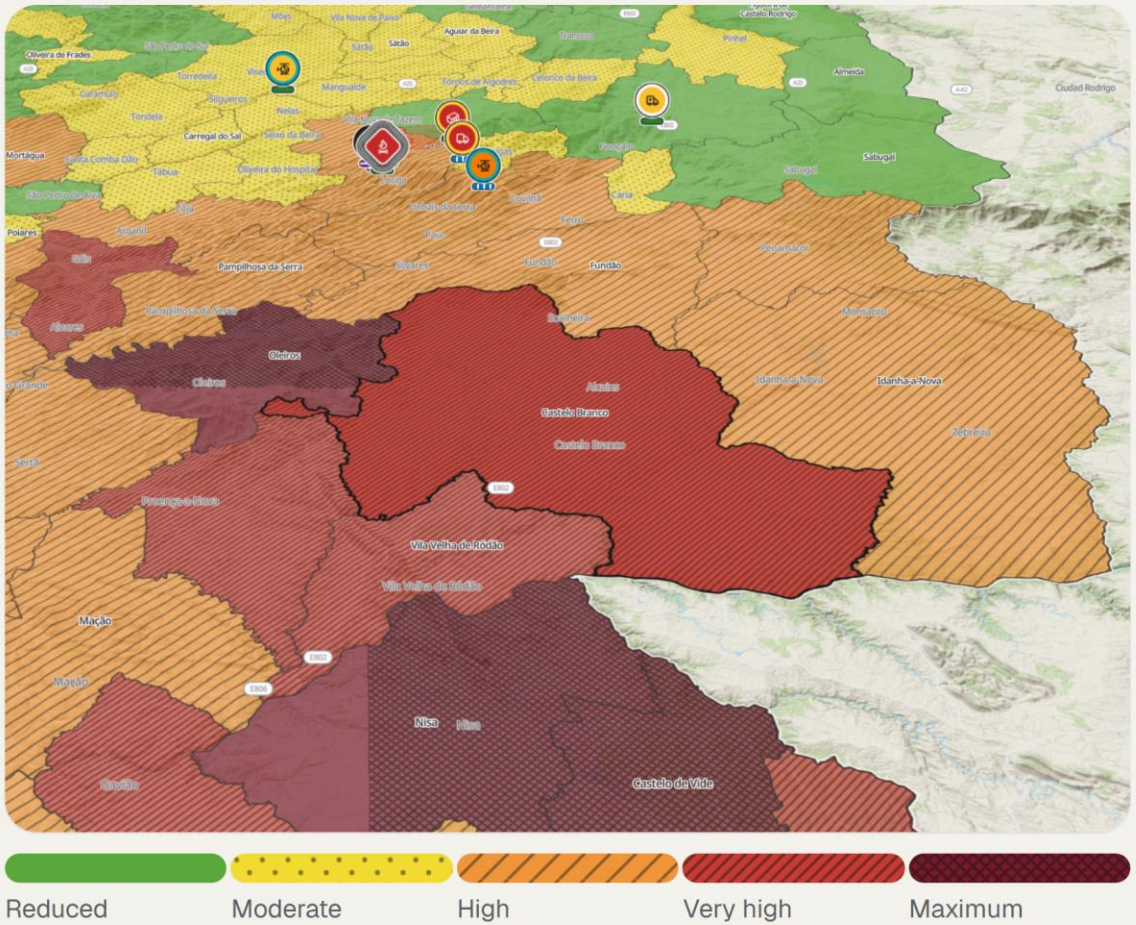
Available · En route · On scene · Unavailable

Ball icon per state



Unacknowledged alert · Overdue order · Orders to send

Risk pattern per level



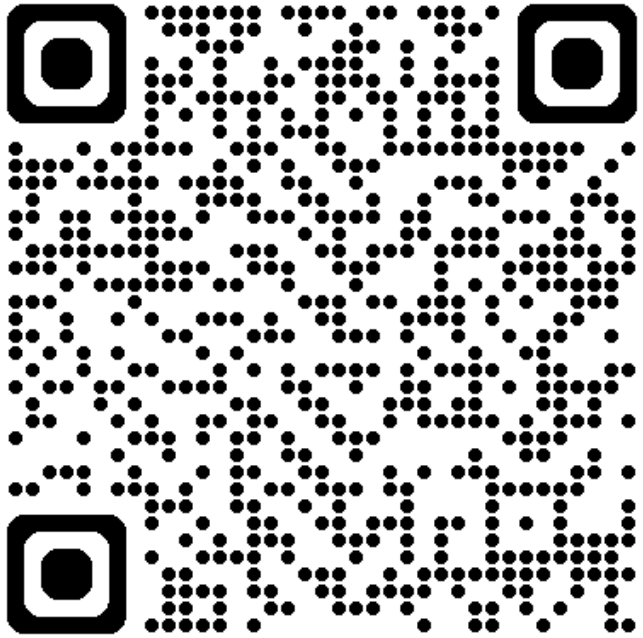
Everything by keyboard · Screen reader · Map as a list · AA contrast, light and dark

UI: On the Dotted Line



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To everything, forever.



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